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(71)**Name of Applicant :**
1)Symbiosis International (Deemed University)
Address of Applicant :Gram: Lavale, Taluka: Mulshi, District: Pune, Maharashtra, India, Pin: 412115 Pune Maharashtra India
(72)**Name of Inventor :**
1)Nimit Prakash
2)Om Narayan Pandit
3)Vraj Patel
4)Vedant Shitole
5)Dr. Shrikrishna Kolhar

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR AUTOMATED PLANT DISEASE DETECTION USING DEEP LEARNING IMAGE CLASSIFICATION A computer-implemented system for automated plant disease detection and classification using deep learning is disclosed. The system comprises a memory storing executable instructions and a processor coupled to the memory configured to execute the instructions. The processor receives plant leaf images captured through user devices via a communication network and preprocesses the images through resizing, normalization, and data augmentation techniques. The system implements a feature extraction module employing MobileNetV2 convolutional neural network architecture with depthwise separable convolutions and inverted residual blocks to extract hierarchical features from the preprocessed images. The extracted features are classified into disease categories using a trained deep learning model generating disease predictions with confidence scores. The system provides real-time plant disease diagnosis on resource-constrained mobile devices with inference latency under 200 milliseconds while maintaining classification accuracy exceeding 80 percent across multiple disease categories. [To be published with Figure 1]

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